SHEBA Tel HaShomer | GLOBAL Healthcare Services

ROM TECHNOLOGIES
MEDICAL, PHYSICAL & TECH PRODUCT DEVELOPMENT

THE JOINT COMMISSION · NATIONAL QUALITY APPROVAL

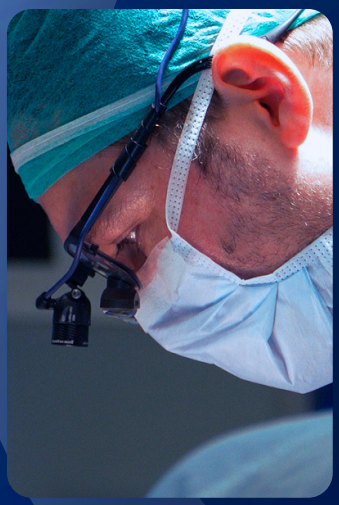
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THE ROM CLINICAL ENGINEERING CENTER

Where clinical insight meets engineering precision.
We transform medical challenges into breakthrough solutions, from initial concept to first-in-human, end-to-end.











Innovation · Research & Development · Clinical Validation · Medical Education · Industry Partnerships

4
Integrated units

3+
Academic institutions

FDA
Aligned regulatory processes

END-TO-END
Concept to commercialization

OUR MISSION

BUILT TO TRANSFORM IDEAS INTO CLINICAL REALITY

The ROM Clinical Engineering Center at Sheba Medical Center brings together innovation, research, and education under one unified framework. We work alongside clinical departments, academic institutions, and industry partners to accelerate the development of medical solutions that genuinely improve patient outcomes.

We believe innovation is not a luxury — it is a clinical and economic imperative. Every project we take on is driven by a real clinical need, guided by rigorous science, and designed with commercial viability in mind.



OUR FOUR PILLARS

A FULLY INTEGRATED INNOVATION ECOSYSTEM

The center is structured around four interconnected units that work as a single, seamless pipeline — from early research to market-ready solutions.

RESEARCH LABORATORIES

01

ENGINEERING IN MEDICINE & PHOTONICS LABS

Two dedicated research laboratories combining applied physics, biomedical engineering, and advanced imaging science to develop the technologies of tomorrow.

Key capabilities

- ECG analysis & cardiac signal processing
- Medical robotics & navigation systems
- 3D visualization — VR & AR environments
- Laser diagnostics & therapeutic photonics
- Biomedical device design & prototyping

ROM DEVELOPMENT CENTER

02

FROM CONCEPT TO FIRST-IN-HUMAN

A dedicated environment for rapid medical device creation, where ideas evolve from early concepts into clinically applicable, first-in-human ready prototypes with precision and speed.

Key capabilities

- Rapid, high-quality prototype engineering
- Design iteration in hospital proximity
- Cross-disciplinary engineering teams
- Timeline & funding strategy support

CLINICAL BETA TRIAL CENTER

03

THE BRIDGE TO CLINICAL ADOPTION

Our gateway for translating innovation into practice. We introduce new technologies into the hospital environment, guide them through regulatory evaluation, and execute clinical trials with care.

Key capabilities

- Startup & industry collaboration
- Regulatory strategy & FDA-aligned compliance
- IRB (ethics committee) submission support
- In-hospital clinical trial execution

EDUCATIONAL CENTER

04

TRAINING THE NEXT GENERATION

A multidisciplinary platform developing the clinician-engineers of tomorrow, in collaboration with leading academic partners across Israel.

Key capabilities

- Reichman University
- Tel Aviv University
- Afeka College of Engineering
- Immersive VR/AR teaching via the Shalev System

WHY PARTNER WITH US

UNIQUE ADVANTAGES YOU WILL NOT FIND ELSEWHERE

HOSPITAL-EMBEDDED ACCESS

Direct access to Sheba's clinical departments, patients, and leading physicians — accelerating validation at every stage of development.

SPEED TO CLINICAL READINESS

Integrated labs, regulatory expertise, and IRB proximity compress timelines significantly compared to standard development pathways.

GLOBAL NETWORK & REACH

Connections with international academic institutions, industry leaders, and technology transfer channels for scale-up and commercialization.





INNOVATION PIPELINE

A STRUCTURED PATH FROM NEED TO MARKET

Every innovation at the center follows a disciplined four-stage pathway — ensuring that only well-validated, clinically sound, and commercially viable solutions advance. We do not simply build devices; we build solutions that reach patients.

1 Concept & Evaluation

- Clinical need identification
- Value & risk analysis
- Feasibility assessment
- Regulatory scoping
- Reimbursement mapping

2 Development

- Prototype design & engineering
- Iterative testing cycles
- Funding strategy
- IP & commercialization planning

3 Validation

- Regulatory preparation
- FDA-aligned documentation
- IRB ethics submission
- In-hospital clinical trials

4 Commercialization

- Technology transfer
- Industry partnerships
- Scale-up support
- Market entry strategy



INNOVATION PIPELINE

TECHNOLOGIES DEVELOPED WITHIN OUR WALLS

A sample of the breakthrough solutions currently in development or already advancing through the clinical pipeline at Sheba Medical Center.



SURGICAL IMAGING
Intraoperative flow imaging system

A novel combination of thermal imaging and laser speckle technology enabling real-time tissue perfusion assessment during breast reconstruction surgery — improving surgical decision-making and patient outcomes.



RECONSTRUCTIVE SURGERY
Advanced breast prosthesis

An innovative prosthetic solution designed to improve functional and aesthetic outcomes for patients following breast surgery, developed in close collaboration with clinical teams.



INFECTION PREVENTION
Antiseptic medical patch

A next-generation wound and access-site management device with targeted antiseptic delivery, designed to reduce healthcare-associated infection rates in clinical settings.



DIAGNOSTIC DEVICES
Oral visualization device

A redesigned laryngoscopy solution offering improved ergonomics, visualization, and usability — addressing unmet needs in both emergency and elective clinical environments.

Additional proprietary solutions are currently in active development across multiple clinical domains. We welcome conversations with partners who wish to explore collaboration opportunities.

MEDICAL ROBOTICS · PROJECT SPOTLIGHT

UR5 ROBOTIC ARM + INTEL REALSENSE D435I

A ROBOT THAT CAN SEE ITS WORKSPACE

A UR5 robotic arm and an Intel RealSense D435i depth camera are connected through a computer. The project focuses on **calibration**: matching the camera's 3D measurements to the robot's coordinates.



THE PURPOSE

Develop a reliable vision-guided robotic platform: see the workspace, understand where objects are, and move with controlled precision.

6	5 kg	850 mm	3D
JOINT AXES	PAYLOAD*	REACH*	DEPTH VISION

*UR5 original-model ratings; confirm model generation on the robot label. Concept imagery is illustrative.

HOW IT WORKS

THREE PARTS. ONE WORKFLOW.

The robot moves, the camera measures the scene, and the computer translates vision data into useful robot commands.



UR5 ROBOT

6 axes · up to 5 kg*

Six-axis arm that positions a tool or gripper and performs controlled movements.

D435I CAMERA

RGB + depth · IMU

Captures colour and depth, helping estimate how far objects are and where they are in 3D.

COMPUTER

Ubuntu 24.04.3 LTS

Runs the vision, calibration and robot-control software that connects the two devices.

THE WORKFLOW

FROM PIXELS TO PRECISE MOTION

1 — CAPTURE

Camera collects depth + colour.

2 — CALIBRATE

Align camera and robot coordinates.

3 — PLAN

Software calculates a target position.

4 — MOVE

Robot performs the commanded motion.

WHY THIS MATTERS FOR MEDICINE

Robotics and computer vision may one day help medical teams with precise positioning, navigation and repetitive tasks. The long-term ambition is to support surgeons during complex procedures — not to claim that this project can currently perform surgery.



Important: the UR5 + D435i setup is an engineering/research platform, not an approved surgical system. It does not replace a doctor. Current robot-assisted surgery systems are operated by trained surgeons.

Specs: Universal Robots UR5 technical sheet · Camera: Intel RealSense D435i · Medical context: U.S. FDA

IMMERSIVE EDUCATION

THE SHALEV SYSTEM — SHEBA'S VIRTUAL LEARNING PLATFORM

The Shalev System redefines how medicine is taught. A centralized control platform synchronizes multiple VR and AR devices simultaneously, enabling a single instructor to guide learners through shared, real-time three-dimensional anatomical environments — regardless of location. It is precise, scalable, and unlike anything currently available in conventional medical education.

WHO IT SERVES

- Medical & engineering students
- Nurses & radiology technicians
- Allied health professionals
- High school & pre-med programs

CORE CAPABILITIES

- Centralized multi-device synchronization
- Shared real-time 3D anatomical models
- Multi-user, multi-location teaching sessions
- Anatomy, pathology & clinical case instruction

ACADEMIC & INSTITUTIONAL PARTNERS

ROOTED IN COLLABORATION

Our educational programs and research activities are conducted in close partnership with leading Israeli academic institutions, alongside international collaborators across academia, industry, and regulatory bodies.

Reichman University | **Tel Aviv University** | **Afeka College of Engineering**

Also working with: Sheba's clinical departments · International research institutions · Global industry & startup ecosystems

"We view innovation not only as a clinical necessity, but as a powerful engine for economic and scientific growth — one that begins with a genuine human need and ends with a solution that reaches patients."

The ROM Clinical Engineering Center — Sheba Medical Center

LET'S BUILD SOMETHING MEANINGFUL

PARTNER WITH THE ROM CENTER

Whether you are a startup at idea stage, a company seeking clinical validation, or an academic institution looking to collaborate – we offer the infrastructure, the expertise, and the hospital proximity to accelerate your work.

Startups & Entrepreneurs

Bring us your idea. We will help you shape it into a viable, validated medical product.

Industry & Corporate

Leverage our clinical infrastructure for validation, trials, and regulatory navigation.

Academic Institutions

Collaborate on research, education programs, and joint innovation initiatives.



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